What Is AI Exposure? Measurement Architecture and Statement-Specific Robustness in Early-Career Employment

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Abstract. Occupational “AI exposure” is not a common empirical object. Prominent indices begin from different technologies, occupational primitives, label sources, aggregation rules, and taxonomies. I trace six defensible architectures through occupational mapping, effective support, a fixed nationally representative employment-stock design, and realized occupational movements. The architectures differ sharply in observable content and rankings. Nevertheless, all six produce negative point estimates for a post-January-2023 high-versus-low early-career employment-stock contrast on literal common support; the confirmatory primary estimate is -0.131 log points. This aggregate directional agreement has clear limits. Among realized occupational switches, 53.3 percent receive opposite directional labels from at least two architectures, although broad occupational assortativity explains nearly all excess conflict relative to a stringent rematching benchmark. A single exploratory joint specification also reveals conditional asymmetry between task-based Eloundou and ability-based AIOE family centroids, but the contrast is sensitive to the distinctive Eloundou alpha measure. Alternative constructions can therefore support the same aggregate employment tail contrast without supporting the same occupational ranking or interpretation. Robustness is statement-specific. The estimates are observational and do not identify causal effects of AI adoption.

JEL codes: J23, J24, O33, C81.

Keywords: artificial intelligence, occupational exposure, measurement, early-career employment, constructed treatments.

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1. Introduction

Economists routinely merge an occupational artificial-intelligence exposure index into worker, firm, or regional data and treat the resulting variable as "AI exposure." Yet the treatment has already been constructed before the regression begins. A researcher has chosen a technology boundary, an occupational primitive, a source of labels, an aggregation rule, a taxonomy bridge, and a support convention. Each choice can change which occupations are called highly exposed, which comparisons remain after conditioning on prior computerization, and which worker movements count as moves toward or away from exposure.

This paper asks a general measurement question: when several defensible constructions exist for the same named treatment, what does it mean for an economic conclusion to be robust? The answer depends on the statement. A high-versus-low comparison consumes treatment-group membership. A continuous regression consumes residual score variation after controls. A mobility claim consumes pairwise occupational rankings. Agreement for one object does not imply agreement for the others. Robustness is therefore not a global property of a variable; it is a property of a specified economic statement and the treatment representation that statement uses.

I study this problem in an empirical laboratory where treatment construction is consequential: the evolution of early-career employment across occupations since the release of ChatGPT. The analysis compares six occupational exposure architectures. Three derive from the ability-based AIOE framework of Felten, Raj, and Seamans (2018, 2021); three derive from the task-based LLM framework of Eloundou et al. (2024). I harmonize each measure to the same occupational taxonomy, document support and residual variation, and then hold the Current Population Survey outcome, age comparison, calendar, estimator, and inference framework fixed while changing the exposure architecture.

The exercise complements rather than competes with outcome-rich and adoption-based studies. Administrative payroll data reveal firm and hiring margins unavailable in the CPS (Brynjolfsson, Chandar, and Chen 2026); linked adoption surveys reveal who uses generative AI and how firms reorganize work (Bick, Blandin, and Deming 2024; Humlum and Vestergaard 2025; Bick et al. 2026). Better outcome data do not eliminate treatment-definition uncertainty. If two architectures classify the same occupation differently, the empirical treatment changes before administrative precision becomes relevant.

Three findings organize the paper. First, the architectures differ materially upstream. AIOE variants are closely related to one another, while direct LLM exposure has substantially different occupational content and rankings. Across 30 exposure-by-computerization constructions, continuous effective residual support ranges from 11.9 to 84.5 equally contributing occupations. Taxonomy repair also changes who enters the estimand: an exact-code AIOE merge onto the post-2018 occupational taxonomy covers only 3.33 percent of computer and mathematical employment, while the documented crosswalk restores 97.7 percent.

Second, the aggregate high-versus-low stock direction is unusually stable. The confirmatory beta-by-Webb coefficient is −0.131 log points: after January 2023, the young-to-older employment-stock ratio evolves 12.3 percent less favorably in the highest than in the lowest exposure quintile. On one literal 444-occupation intersection, all six architecture-specific point estimates are negative. This is point-estimate sign robustness, not a common causal parameter. One interval includes zero, and a predeclared simultaneous one-sided confidence-band criterion does not establish that all six coefficients are negative at familywise 95 percent confidence.

Third, occupational ranking and conditional family content are less interchangeable. Among realized occupational switches with all six scores, 53.28 percent receive at least one positive and one negative directional label. That disagreement is frequent, although a stringent benchmark that preserves broad origin-destination assortativity raises expected conflict to 52.32 percent, leaving only a 0.96 percentage-point realized excess. In one predeclared exploratory continuous model, a family-balanced consensus coordinate is negative and the AIOE-versus-Eloundou disagreement coordinate is positive. Exact re-expression in the original family basis places the negative conditional association more heavily on the Eloundou centroid. The contrast is stable to deleting any single occupation but sensitive to the distinctive Eloundou alpha measure, so it is evidence about the frozen construction rather than a universal family ranking.

These results contribute a linked account of constructed-treatment uncertainty. Existing comparisons show that exposure indices differ (Gimbel, Kendall, and Kulsakdinun 2026; Merola et al. 2026; Yin, Vu, and Persico 2026; Yin and Ogut 2026). This paper follows those differences through taxonomy, support, estimator information, same-design employment contrasts, and mobility labels. The distinction between treatment-group membership, residual continuous variation, and pairwise ranking explains how aggregate directional robustness can coexist with downstream non-interchangeability.

The paper proceeds as follows. Section 2 relates the analysis to exposure measurement, technology and labor, and generated covariates. Sections 3–5 describe the six architectures, CPS design, and support diagnostics. Sections 6 and 7 report stock and mobility evidence. Section 8 addresses computerization, remote work, dynamics, and the limits of mechanism claims. Sections 9 and 10 draw implications for research using constructed treatments.

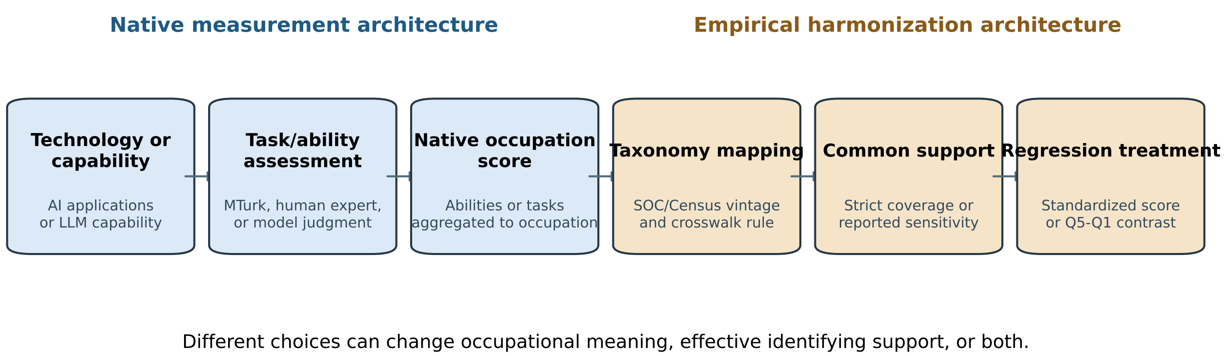

FIGURE 1. From native measurement architecture to regression treatment

Notes: The figure separates choices made by an exposure measure's authors from harmonization choices made by downstream researchers. Each stage can alter occupational meaning, empirical support, or both. Exposure is a measure of potential technological susceptibility, not observed adoption.

2. Related Literature

2.1. Occupational exposure and labor-market measurement

The first literature constructs occupational exposure to technological capabilities. AIOE links progress in ten AI applications to 52 O*NET abilities and then combines application-ability relatedness with occupational ability requirements (Felten, Raj, and Seamans 2018, 2021). Eloundou exposure instead asks whether a large language model, alone or with complementary software, can reduce the time required for occupational tasks by at least half (Eloundou et al. 2024). Webb (2020) uses textual overlap between patents and occupational tasks to distinguish software, robots, and AI. The OECD's capability-gap framework provides a further multidimensional construction (Organisation for Economic Co-operation and Development 2026). These measures encode distinct technological primitives; none directly observes firm use.

Recent work makes instability across measures an object of study. Yin, Vu, and Persico (2026) holds an LLM rubric fixed and changes the annotating model, producing substantial score and downstream coefficient variation. Yin and Ogut (2026) shows that platform-log measures partly reflect the occupational composition of platform users. Cross-index reviews similarly emphasize that exposure is susceptibility rather than an employment prediction (Gimbel, Kendall, and Kulsakdinun 2026; Merola et al. 2026). The present paper does not claim the first comparison of exposure scores. Its contribution is to connect construction to taxonomy, residual support, estimator information, common-support inference, and economic statements that consume different representations of the score.

Studies with adoption or richer outcomes answer complementary questions. Hampole et al. (2025) link task exposure and firm adoption to labor demand and reallocation. Humlum and Vestergaard (2025) combine Danish adoption surveys with administrative records and find task reorganization alongside precise near-term earnings and hours nulls. Bick, Blandin, and Deming (2024) measure adoption in a nationally representative survey, while Bick et al. (2026) link reported use to occupations and tasks and show that exposure explains only part of adoption. Pulito et al. (2026) also document technology-specific adoption patterns. These papers improve observation of use. The present design isolates uncertainty in how occupations are assigned a potential-exposure treatment.

2.2. Computerization, tasks, and competing shocks

The task approach to technological change distinguishes substitution within tasks from changes in the composition and demand for work (Autor, Levy, and Murnane 2003; Autor and Dorn 2013; Acemoglu and Restrepo 2019). That distinction motivates conditioning AI exposure on multiple measures of prior computerization rather than treating them as interchangeable. Webb software-patent exposure, O*NET computer-use importance and level, routine-task intensity, and automation susceptibility remove different occupational comparisons. Their variation is part of the estimand.

Remote work provides a separate competing interpretation. Dingel and Neiman (2020) measure occupational feasibility of working from home, while Emanuel, Harrington, and Pallais (2026) connect proximity to feedback and skill development and document weaker post-pandemic outcomes for young college graduates in remotable work. Their individual unemployment outcome and age groups differ from the stock contrast here. Remotability is therefore a substantive robustness margin, not a variable that establishes whether AI or remote work caused the observed pattern.

2.3. Constructed and generated covariates

Classical errors-in-variables intuition is incomplete when measurement errors are systematic, architecture-dependent, and rank-changing. Multiple-proxy methods require additional exclusion and measurement assumptions (Hu and Schennach 2008). Recent work shows that plugging AI- or machine-learning-generated covariates into downstream regressions can invalidate ordinary inference and generally requires validation structure for correction (Battaglia et al. 2024; Christensen and Hansen 2026; Duan and Pelger 2026; Ludwig, Mullainathan, and Rambachan 2026). I do not observe validated true exposure, so I do not estimate a corrected latent treatment. Instead, I make construction choices observable and ask which claims survive alternatives.

Composite-index research supplies a useful analogy: weights, normalization, scope, and missing-component rules can alter ranks even when aggregate correlations are high (Decancq and Lugo 2013). Here each architecture is treated as a distinct empirical treatment rather than a specification selected by outcome fit. Common-multiplier procedures preserve cross-architecture covariance, and simultaneous inference is tied to the exact compound statement being evaluated (Romano and Wolf 2005).

3. Exposure Architectures and Harmonization

3.1. Six constructions

Table 1 describes the six exposure architectures. The three AIOE variants share application-to-ability links but differ in how scores are carried from source occupations to the target occupational taxonomy. Administrative equal aggregation gives source occupations equal weight; the ability-direct construction rebuilds the index from target O*NET ability requirements; the source-weighted construction uses May 2018 OEWS employment weights. The three Eloundou variants share GPT-4 task labels but change capability scope. Alpha contains direct LLM acceleration (E1), beta adds half weight to software-complemented tasks (E2), and broad exposure includes E1 and E2 fully.

TABLE 1. Six occupational exposure architectures

Measure	Native primitive	Aggregation distinction	Final representation
AIOE administrative	Ten AI applications mapped to 52 occupational abilities	Equal administrative aggregation across source occupations	Census-2018 score; weighted quintiles or continuous z-score
AIOE ability	Same application-to-ability links	Direct reconstruction from target O*NET ability requirements	Same
AIOE source-weighted	Same application-to-ability links	May 2018 OEWS source-employment weights	Same
Eloundou alpha	Direct LLM task acceleration	Share of tasks classified E1	Same
Eloundou beta	LLM plus limited software complementarity	E1 plus one-half of E2	Same
Eloundou broad	LLM plus software	E1 plus E2	Same

Notes: The table describes construction rather than ranking measure quality. All measures are harmonized to Census-2018 occupations through documented official bridges. The primary support rule excludes partially scored target occupations instead of renormalizing observed components. Exposure measures potential technological susceptibility, not observed adoption.
Source: Authors' reconstruction of Felten, Raj, and Seamans (2018, 2021) and Eloundou et al. (2024).

These are not six noisy measurements of a known common scalar. The ability-based family measures broad applicability of AI capabilities; the task-based family measures a particular time-saving rubric for LLMs and complementary software. A common latent component may exist, but identifying it requires assumptions not supplied by the six scores. The analysis therefore compares observable architectures and avoids designating a preferred measure based on outcomes.

3.2. Taxonomy mapping and support

Raw CPS occupations use Census 2010 codes through 2019 and Census 2018 codes thereafter. Pre-2020 records are route-expanded using official Census conversion rates. SOC-based exposure measures are collapsed at six digits and carried through documented SOC-to-Census bridges using source- or target-vintage weights as appropriate. The lineage fixes tie handling, missing components, and normalization.

This mapping is economically consequential. A literal exact-code merge between AIOE's SOC-2010 publication taxonomy and post-2018 outcome taxonomies misses split and renumbered occupations. In the computer and mathematical group, exact matching covers 3.33 percent of 2021 OEWS employment; the documented repair covers 97.7 percent. A four-row decomposition separates score correction from sample composition. Repairing scores on the original support changes the continuous coefficient only from −0.01885 to −0.01920. Expanding support changes it to −0.03156; removing computer and mathematical occupations gives −0.02940. The principal mapping effect is admission to the estimand, not a large revision among occupations already matched.

The primary support rule is strict no-renormalization. Every mapped source component must have a finite score; partially scored occupations are excluded rather than reconstructed by rescaling observed components. Two alternative rules are retained in the appendix. Literal common support is the 444-occupation intersection across all six strict-support measures and represents 83.14 percent of model-period employment. Fixing occupation support does not fix scores, ranks, or quintile membership.

3.3. Consensus and disagreement coordinates

To organize the six measures without selecting among them, each is standardized using frozen employment weights. Let A_O denote the equal-weight centroid of the three AIOE measures and E_O the equal-weight centroid of the three Eloundou measures. Define

$$F_O = \frac{A_O + E_O}{2}, \qquad G_O = \frac{A_O - E_O}{2}. \tag{1}$$

F is a family-balanced consensus coordinate; G is the AIOE-versus-Eloundou disagreement coordinate. Neither is a recovered true exposure index. Their use was limited to one predeclared exploratory outcome model after treatment-side construction and stopping rules were recorded.

4. CPS Data and Empirical Design

4.1. Employment stocks

The analysis uses 9,262,480 IPUMS CPS records and aggregates employed respondents ages 22–65 with the basic monthly final weight into occupation-by-age-group-by-month stocks. Young workers are ages 22–25; the comparison group is ages 26–65. The extract spans January 2017 through July 2026. March is absent in 2017–2021 and October 2025 is absent, leaving 109 observed months. December 2022 is a transition month excluded from static models. January 2023 through July 2026 contains 42 observed post months.

The outcome is a stock, not an individual employment probability. It can move through entry, employment exit, occupational switching, or changes in the older comparison stock. The design consequently estimates a young-relative occupational employment gradient; it does not by itself identify displacement, hiring, or separation.

4.2. Headline estimator

Let N_{oat} be the CPS-weighted employment stock in occupation o, age group a, and observed month t. Let $Young_a$ equal one for ages 22–25, $Post_t$ equal one from January 2023 forward, and Q_{oq} denote architecture-specific exposure quintile q, with Q1 omitted. With standardized Webb software exposure W_o, the implemented conditional mean is

$$E[N_{oat} \mid X] = \exp\Bigg[\gamma_{oa} + \delta_{ot} + \lambda_{at} + \sum_{q=2}^{5} \beta_q Q_{oq} Young_a Post_t + \theta W_o Young_a Post_t\Bigg]. \tag{2}$$

The fixed effects are occupation-by-age group, occupation-by-month, and age-group-by-month. CPS person weights construct N_{oat}; no second survey weight is applied to the cell likelihood. The code estimates the algebraically equivalent grouped-binomial conditional likelihood for the young share within occupation-month. Inference clusters by occupation and uses the frozen 999-draw one-step Rademacher wild-score procedure.

The target β_5 is the post-period change in the young-to-older stock ratio for Q5 relative to Q1, conditional on Q2–Q4 and Webb. It is not a continuous exposure slope. Every architecture defines its own employment-weighted quintiles, so the same coefficient label can refer to different occupation sets. The paired beta-minus-alpha comparison holds the support, outcome, estimator, computerization control, and bootstrap draws fixed but still compares architecture-specific Q5/Q1 treatments.

4.3. Design status and interpretation

The primary alpha/beta architecture grid, support rules, estimator, and inference were fixed before protected post-period outcomes were opened. Common-support six-measure models, estimator-information diagnostics, the age profile, longitudinal flows, reallocation comparisons, and the F/G model are explicitly exploratory. This status distinction governs claims rather than determining which estimates are displayed.

All estimates are observational. Exposure is not adoption, the January 2023 indicator is not an instrument, and occupation-level treatment is correlated with other shocks. The identifying comparison is descriptive conditional evolution across occupational exposure groups. The design is useful for testing stability across constructed treatments without licensing causal attribution to AI.

5. Architecture Divergence and Empirical Support

5.1. Occupational content and rankings

The three AIOE variants correlate between 0.981 and 0.996 in employment-weighted levels and between 0.982 and 0.993 in ranks. Beta and broad Eloundou exposure are also closely related. Cross-family similarity is less uniform: the weighted level correlation between ability-direct AIOE and alpha is 0.276, and their weighted rank correlation is 0.258. Pairwise Q5 overlap between AIOE and alpha is only 0.216–0.237; overlap between AIOE and beta is 0.414–0.465.

Observable content differs in parallel. The AIOE variants load strongly on cognitive content, education, wages, teleworkability, and computer use. Alpha has weaker relationships with those characteristics; beta and broad are intermediate. These are descriptive projections, not validity tests against a true exposure label.

5.2. Residual treatment and estimator information

Nominal occupation count is not effective support. For each exposure-by-computerization architecture, I residualize the standardized exposure on the selected computerization measure and form employment-weighted residual-variance shares s_o. The effective count is

$$N_{\text{eff}} = \frac{1}{\sum_o s_o^2}. \tag{3}$$

Across 30 constructions, N_{eff} ranges from 11.9 to 84.5. Alpha conditional on Webb has 17.4 effective occupations and a top-five share of 41.6 percent; beta conditional on Webb has 53.3 and 22.2 percent.

The categorical estimator consumes a different object. Absorbing the exact Q5/Q1 design under fitted information weights produces 42.9–71.1 effective occupations across 12 headline models. For beta/Webb, continuous and estimator-specific effective counts are 53.3 and 43.3; for alpha/Webb they are 17.4 and 56.1. Continuous residual support and fitted conditional information are related but not interchangeable, and neither is a realized deletion diagnostic.

TABLE 2. Treatment content and effective support

Exposure architecture	R^2 on eight characteristics	Residual SD	Continuous effective occupations	Top-five share
AIOE administrative	0.964	0.189	35.5	30.4%
AIOE ability	0.954	0.214	37.8	29.0%
AIOE source-weighted	0.971	0.171	44.2	24.9%
Eloundou alpha	0.368	0.795	31.5	33.7%
Eloundou beta	0.743	0.507	36.9	28.0%
Eloundou broad	0.811	0.435	31.5	29.7%
Estimator-specific bridge (selected architectures)				
Beta / Webb			53.3 / 43.3	22.2% / 24.6%
Alpha / Webb			17.4 / 56.1	41.6% / 21.3%

Notes: The upper panel residualizes each standardized exposure on eight preselected occupational characteristics using frozen pre-period employment weights; the effective count is the inverse Herfindahl index of employment-weighted residual-variance shares. The lower panel reports continuous residual-treatment support followed by estimator-specific conditional-information support for the categorical stock estimator. These are descriptive support diagnostics, not causal influence measures. The stock estimator uses 468 occupations, ages 22–65, and 108 static months; the conditional-information calculation is exploratory and uses fitted information weights.

The support audit changes how coefficient robustness should be read. Similar coefficients can arise from treatments with different ranks and residual comparisons. Conversely, a concentrated continuous residual may not imply a concentrated categorical estimator. Reporting architecture, support, and fitted information alongside coefficients makes these distinctions visible without treating any diagnostic as a causal leverage decomposition.

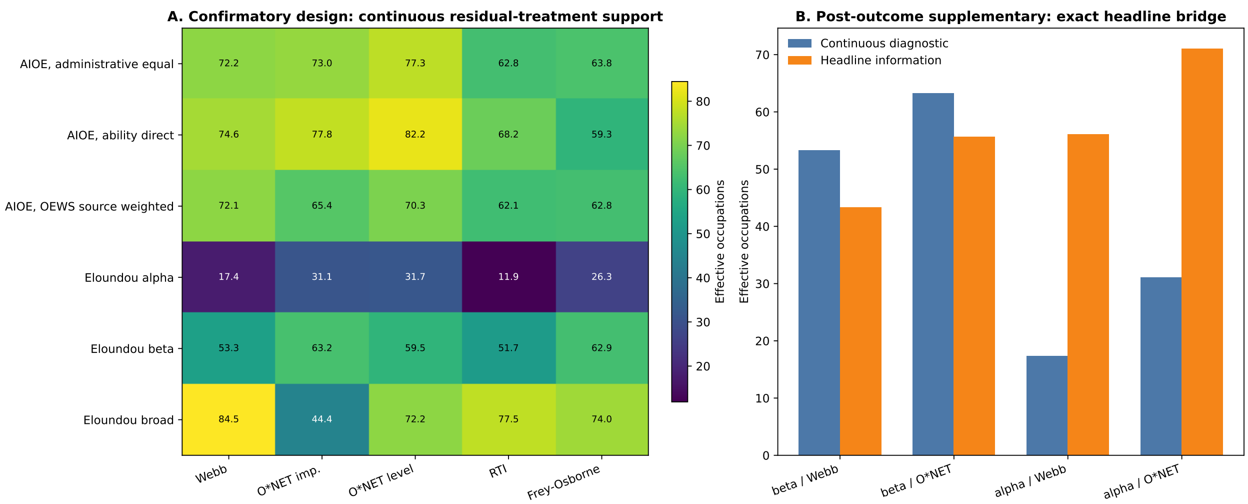

FIGURE 2. Continuous residual-treatment support and estimator-specific information

Notes: Panel A reports the inverse-Herfindahl effective count of employment-weighted residual-treatment variance across six exposure measures and five prior-computerization controls. Panel B compares that continuous diagnostic with fitted conditional-information support for four categorical headline estimators. The diagnostics describe different objects and are not leave-one-occupation influence measures.

6. Employment-Stock Results

6.1. Confirmatory tail contrasts

Across the 12 prespecified alpha/beta architectures, Q5-versus-Q1 estimates range from −0.0971 to −0.2085 log points. The primary strict-support beta-by-Webb estimate is −0.1311 (occupation-cluster SE 0.0444; wild-score 95 percent interval [−0.2170, −0.0451]; $p = .003$) across 468 occupations. The transformation $100[\exp(\widehat{\beta}_5) - 1]$ is −12.3 percent. This is less favorable evolution of the young-to-older employment-stock ratio in the high-exposure tail, not an individual job-loss probability.

The intermediate quintiles are not monotone. Relative to Q1, the primary Q2–Q5 coefficients are −0.0855, −0.0478, −0.0970, and −0.1311. The paper therefore reports a tail contrast, not an exposure dose-response. An existing exploratory age-profile model places negative point estimates in the 18–21 and 22–25 groups, with both intervals including zero, while ages 26–30 have a small positive estimate with an interval spanning zero; this pattern supports "early-career" only descriptively.

A fixed-treatment deletion audit clarifies occupation influence. Deleting one occupation at a time while preserving full-sample scores, quintiles, Webb scaling, calendar, and specification yields estimates from −0.14238 to −0.11055. No deletion changes the sign. Most movements are small: the employment-weighted median is −0.00001, and the 5th and 95th percentiles are −0.00817 and +0.00377.

Influence appears at both tails. Fast food and counter workers are in the frozen beta Q1 reference tail; deleting them moves the coefficient from −0.13107 to −0.11055. Thus, part of the magnitude reflects favorable young-relative evolution at the low-exposure tail, not only deterioration at the high-exposure tail. Customer service representatives are in Q5; their deletion moves the estimate to −0.11311. Software developers are also in Q5 but rank tenth in absolute deletion influence. Deleting them makes the coefficient slightly more negative, −0.13879, a movement of −0.00771. The result is therefore not a software-developer artifact. These deletions describe coefficient influence; they do not identify the economic shocks behind any occupation's evolution.

6.2. Literal common support and the compound sign statement

Figure 3 holds occupational support fixed at 444 occupations. The three AIOE coefficients are −0.0739, −0.1029, and −0.1021; the alpha, beta, and broad Eloundou coefficients are −0.1013, −0.1290, and −0.1465. All point estimates are negative. The administrative-AIOE

TABLE 3. Early-career employment-stock contrasts

Status	Exposure architecture	Occupations	Coefficient	95% interval	p
Confirmatory	Beta / Webb, primary support	468	-0.1311	[-0.2170, -0.0451]	.003
Exploratory	AIOE administrative, common support	444	-0.0739	[-0.1492, 0.0014]	.057
Exploratory	AIOE ability, common support	444	-0.1029	[-0.1772, -0.0285]	.005
Exploratory	AIOE source-weighted, common support	444	-0.1021	[-0.1804, -0.0238]	.014
Exploratory	Eloundou alpha, common support	444	-0.1013	[-0.1855, -0.0172]	.018
Exploratory	Eloundou beta, common support	444	-0.1290	[-0.2161, -0.0418]	.004
Exploratory	Eloundou broad, common support	444	-0.1465	[-0.2329, -0.0602]	.003
Exploratory	Consensus F, continuous per SD	444	-0.0404	[-0.0693, -0.0115]	.007
Exploratory	Disagreement G, continuous per SD	444	+0.0309	[0.0018, 0.0600]	.040

Notes: The outcome in the first seven rows is the occupation-by-age employment stock. The coefficient is the post-January-2023 change in the young-to-older stock ratio in architecture-specific Q5 relative to Q1, conditional on Q2–Q4, occupation-by-age, occupation-by-month, and age-by-month fixed effects, and standardized Webb software exposure. Common support is the same literal 444-occupation intersection, but each architecture defines its own quintiles. The final two rows are continuous coefficients from one predeclared exploratory model containing standardized F, G, and Webb jointly; they are a different estimand. Intervals and p-values use the existing 999-draw occupation-cluster wild-score procedure.

interval narrowly includes zero.

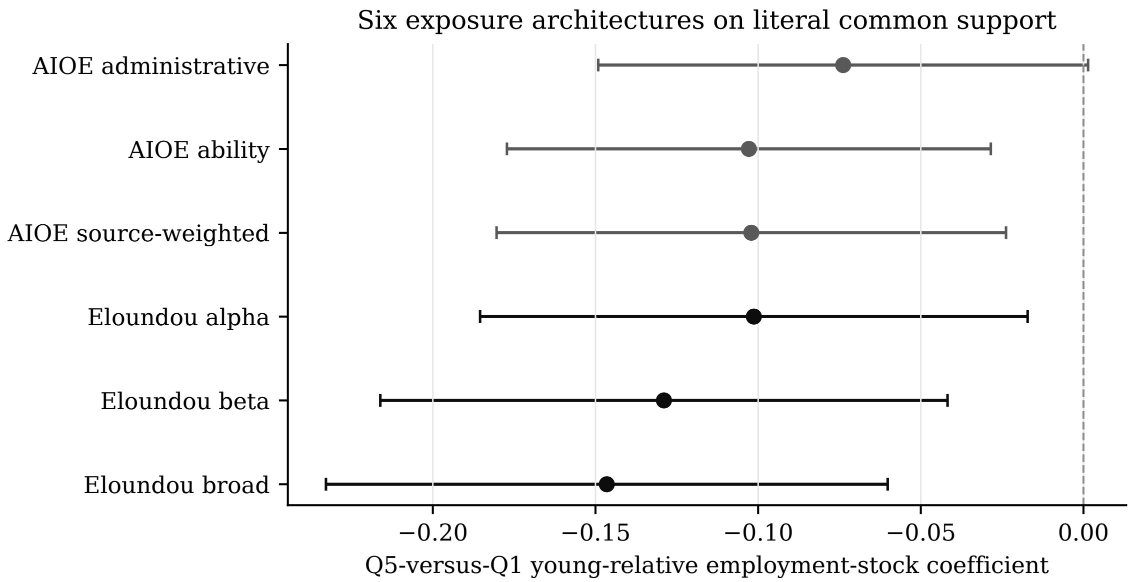

FIGURE 3. Common-support employment-stock coefficients across six architectures

Notes: Points are architecture-specific Q5-versus-Q1 young-relative employment-stock coefficients estimated on the identical 444-occupation intersection. Lines are existing 95 percent one-step occupation-cluster wild-score intervals. The sample, outcome, fixed effects, Webb control, and inference procedure are held fixed; exposure scores, ranks, and quintile membership vary by architecture. All models are exploratory common-support comparisons.

Point-estimate sign robustness and simultaneous inference answer different questions. Under the predeclared common-multiplier max-statistic procedure, the one-sided upper bound for administrative AIOE remains positive. The specific familywise 95 percent simultaneous-band criterion therefore does not establish that all six coefficients are negative at once. This statement does not undo the descriptive fact that all six estimates are negative, nor does it assert that the six models estimate one common parameter.

The paired beta-minus-alpha contrast on the 468-occupation support is −0.0324 log points (paired SE 0.0370; 95 percent interval [−0.1023, 0.0376]; $p = .403$). The design does not detect a difference; an interval containing zero does not establish economic equivalence.

Prospective simulations materially overstated realized precision. The realized-to-prospective SE ratio is 3.65 for the primary coefficient (0.04441 versus 0.01217) and 3.17 for the paired contrast (0.03697 versus 0.01167). The simulations preserved the observed exposure distribution, weights, design matrix, and ordered historical residual paths, but could not reproduce the realized post-period process. Code-history audits do not isolate a unique cause. The old power and MDE calculations are design-history provenance, not evidence about realized power, and they do not change realized intervals or p-values.

6.3. Consensus and between-family evidence

The single exploratory joint continuous model estimates Equation 2's young-by-post slope structure with standardized F, standardized G, and standardized Webb entered together. The family-balanced consensus coefficient is $b_F = -0.04036$ per SD (wild-score interval $[-0.06925, -0.01147]$; $p = .007$). Conditional on F and Webb, the family-disagreement coefficient is $b_G = +0.03089$ (interval $[0.00180, 0.05999]$; $p = .040$). A max-$|t|$ test of both coefficients equal to zero gives $p = .008$. Separately, an exploratory F-quintile model reproduces the negative aggregate tail direction at -0.1285.

The same fitted two-dimensional model can be expressed in the original family-centroid basis. Because F and G were standardized separately, scale-correct algebra—not simple addition and subtraction—is required. Per weighted SD, the implied AIOE-centroid coefficient is $+0.02493$, while the implied Eloundou-centroid coefficient is -0.06148. The transformed predictor reproduces the original F/G predictor to machine precision. This is an exact change of basis of the same model, not a second regression or corroborating evidence.

Inference differs across transformed objects. The original-unit AIOE-minus-Eloundou contrast is exactly a positive rescaling of G and therefore inherits G's existing wild-score interval $[0.00559, 0.18621]$ and $p = .040$. The individual AIOE and Eloundou coefficients combine F and G. Their intervals, $[-0.01492, 0.06477]$ and $[-0.10836, -0.01460]$, use the serialized common-draw covariance and a normal 1.96 critical value because draw-level F/G shifts were not retained. They are not wild-score intervals.

Two audits bound the interpretation. First, deleting any one of the 444 occupations leaves G positive, between 0.02513 and 0.03599; the largest movement is 0.00577 after deleting driver/sales workers and truck drivers. The coefficient is not a single-occupation artifact. Second, changing the treatment-side construction is more consequential: removing alpha lowers the level and rank correlations with the frozen G to 0.865 and 0.855 and changes the zero-direction classification for 17.9 percent of employment. No leave-one-measure labor-outcome models were run. The defensible conclusion is narrow: in this exploratory continuous joint specification, the negative conditional association loads more heavily on the Eloundou-family centroid, but that family coordinate is materially influenced by alpha. The result neither identifies a causal mechanism nor determines which exposure family is correct.

7. Occupational Reallocation and Ranking Interchangeability

7.1. Adjacent-month construction

The CPS rotation design permits exact adjacent-month links for origins in months-in-sample 1, 2, 3, 5, 6, and 7. The eight-month gap from month-in-sample 4 to 5 is excluded. Origin longitudinal weights are used, and December 2019 to January 2020 is excluded when destination occupation matters because the coding taxonomy changes. Dependent interviewing improves consistency for eligible continuing interviews without eliminating response or coding error (U.S. Census Bureau 2006, 2025; Mellow and Sider 1983; Kambourov and Manovskii 2008). A persistence check requires a legitimate third observation and an A-to-B-to-B path.

7.2. Directional disagreement

For each observed occupational switch, an architecture assigns a positive, negative, or zero direction from the destination-minus-origin exposure score. On the 108,500 switches with finite scores under all six architectures, 53.28 percent receive at least one positive and one negative label; 45.56 percent receive the same nonzero direction under all six. Pairwise official-weight Cohen κ ranges from 0.125 to 0.932. Moving from six-way to pair-specific support changes weighted agreement by at most 1.95 percentage points. The result is not created by restricting attention to the literal six-way intersection.

The classification statistic is intentionally permissive: any sign reversal counts, however small the movement. It therefore measures non-interchangeability of directional rankings, not economically large displacement. The young-by-post difference in conflict is essentially zero, with a 95 percent interval of $[-2.67, 2.66]$ percentage points. Architecture disagreement is a general feature of the ranking, not one detected specifically among young movers after 2022.

7.3. A harder benchmark

Independent age-by-month rematching implies 45.27 percent conflict, 8.02 points below the realized rate. That benchmark ignores occupational assortativity. A harder procedure preserves age-by-month, broad origin family, broad destination family, and the detailed origin and destination marginals within each stratum while destroying only the observed detailed pairing. It covers 98.31 percent of switch weight. Under this benchmark, expected conflict is 52.32 percent, leaving a realized excess of 0.96 points—below the predeclared

TABLE 4. Directional agreement in realized occupational switches

Descriptive object	Result
Six-way comparable switches	108,500
At least one opposite directional label	53.28%
Same nonzero direction under all six measures	45.56%
Pairwise official-weight Cohen κ range	0.125–0.932
Maximum pair-support change in weighted agreement	1.95 pp
Independent-rematching conflict benchmark	45.27%
Broad-assortative-rematching conflict benchmark	52.32%
Realized excess over broad benchmark	0.96 pp

Notes: The sample consists of valid adjacent-month employed-to-employed CPS links with finite scores under all six architectures. A switch is conflicting when at least one exposure score rises and another falls; any nonzero sign reversal counts irrespective of magnitude. Official origin longitudinal weights are used. The hard benchmark preserves age-by-month and broad origin-destination occupational-family assortativity while rematching detailed occupations. The 0.96-point gap is descriptive and below the predeclared one-point meaningful-gap threshold. These statistics concern ranking interchangeability, not displacement or worker response to AI.

one-point meaningful-gap threshold.

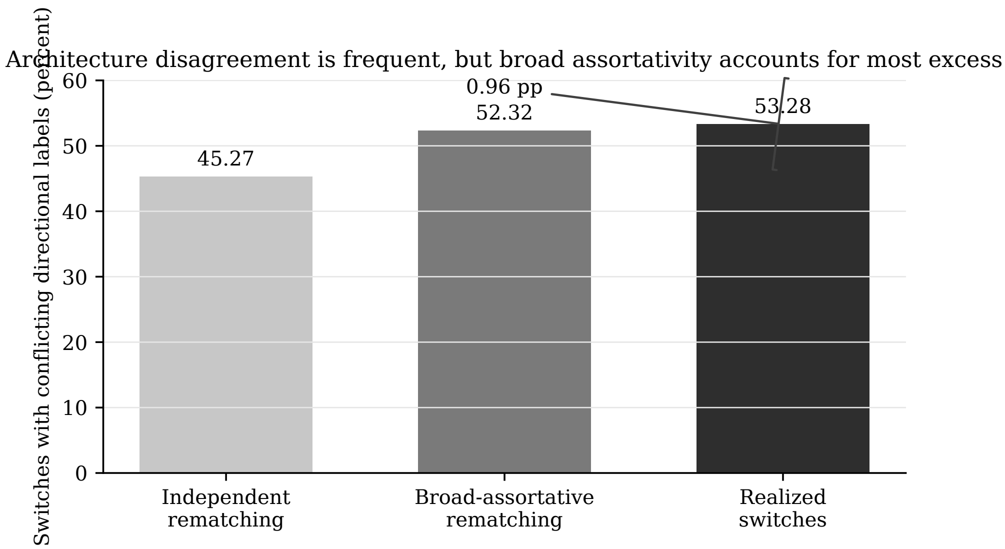

FIGURE 4. Realized directional conflict and rematching benchmarks

Notes: Conflict is the official-weight share of realized adjacent-month switches for which at least one of six exposure architectures rises and another falls. The independent benchmark rematches occupations within age-by-month cells. The broad-assortative benchmark additionally preserves broad origin-destination occupational-family structure and detailed marginals. The 0.96-point realized excess is descriptive, not a conventional sampling estimate.

The combined conclusion is deliberately two-sided. Architecture disagreement is frequent: researchers can label the same occupational movement in opposite directions. But actual worker pairings are not meaningfully unusually conflict-heavy once broad assortativity is preserved. A secondary pattern in which conflict falls from 94.59 to 19.06 percent across bins of absolute consensus movement is reported only in the appendix because its connection to the same component used to define disagreement is partly mechanical.

7.4. Flow margins

An exploratory longitudinal exercise separately estimates employment exit, occupational outflow among linked employed workers, and the destination composition of entries from nonemployment. The coefficients are 0.1195, 0.0107, and −0.0888, respectively, and all intervals include zero. The risk sets and parameters differ and do not add to the stock coefficient. The CPS evidence therefore does not precisely distinguish exit, outflow, or entry destination as the mechanism behind the stock pattern; detailed estimands, weighting, coding checks, and persistence results appear in the online appendix.

8. Computerization, Remote Work, and Competing Interpretations

The primary design conditions beta exposure on Webb's software-patent measure. Substituting other pre-existing technology measures changes the remaining occupational comparison. The beta coefficient is −0.2085 with O*NET computer-use importance, −0.1512 with O*NET computer-use level, −0.1277 with routine-task intensity, and −0.1001 with Frey–Osborne automation susceptibility, compared with −0.1311 under Webb. All corresponding intervals exclude zero, but the more than twofold coefficient range rules out the claim that "controlling for computerization" is a single invariant operation.

Remotability is strongly correlated with several exposure measures, especially AIOE, but correlation creates imprecision rather than automatic non-identification. Adding Dingel–Neiman occupational work-from-home feasibility leaves the beta estimate negative at −0.1000 with interval [−0.1902, −0.0099]. Under alpha, the estimate is −0.0786 with interval [−0.1550, −0.0021]. The remote coefficient changes sign across the two architectures. The narrow conclusion is that occupational remotability does not mechanically absorb the exposure tail contrast in these models. The exercise does not show that AI dominates remote work or that remote work is unimportant.

The categorical event study reports no detected differential pretrend under the existing common-multiplier max-statistic criterion: the largest absolute pre-period t statistic is 1.502 and the wild-score p value is .929. Forty of 42 observed post coefficients are negative, but the path is neither immediate nor monotone. A static post coefficient is an era average. Interest rates, technology-sector adjustment, return-to-office changes, post-pandemic normalization, and occupation-by-age shocks remain plausible concurrent forces.

The occupation-influence audit reinforces this limitation. Part of the extreme-tail contrast is supplied by favorable young-relative evolution in a low-exposure food-service occupation. This is consistent with the need to interpret the coefficient against broader post-pandemic labor-market normalization, but the deletion result does not identify normalization—or any other mechanism—as its cause.

These competing interpretations define rather than defeat the paper's contribution. The empirical object is a conditional association whose stability is examined across constructed exposure treatments. The analysis can establish that a sign persists across specified architectures, that magnitudes depend on conditioning choices, and that occupational rankings differ. It cannot convert exposure into adoption or separate AI causally from every contemporaneous shock.

9. Implications for Research with Constructed Treatments

The first implication is practical: an exposure coefficient should be reported as an architecture, not merely a variable name. Technology scope, occupational primitive, label source, aggregation rule, taxonomy bridge, support convention, normalization, and treatment representation jointly define the empirical object. A replication package should preserve these decisions alongside the regression code.

Second, common support solves composition but not treatment identity. Restricting all measures to the same occupations prevents missingness from mechanically driving coefficient differences. It does not make scores, ranks, quintiles, residual variation, or fitted information equal. Continuous correlations, rank overlap, residual support, estimator information, and deletion influence answer distinct questions.

Third, robustness must be attached to a statement. Here the negative aggregate tail direction survives broad architecture changes. Coefficient magnitudes and conditioning on prior computerization vary. Marginal occupational movements are frequently assigned opposite signs. A family-basis decomposition contains conditional asymmetry but is sensitive to one component measure. These facts coexist because they consume different representations of the constructed treatment.

Fourth, constructed covariates invite uncertainty beyond conventional regression inference. The wild-score intervals condition on the selected exposure architecture; they do not account for the process that generated labels or for uncertainty in taxonomy mapping. Without validation data, the analysis cannot estimate a latent correct exposure or a full measurement-error correction. Transparent architecture comparison is therefore a complement to, not a substitute for, methods that use validation structure.

The logic extends beyond AI. Trade exposure, routine-task intensity, climate risk, policy intensity, and other generated indices combine conceptual and mechanical choices before estimation. For any such treatment, researchers should ask which observations supply variation and which economic statement consumes the resulting representation. A coefficient robust to one change does not validate every ranking or interpretation derived from the index.

10. Conclusion

Six defensible occupational AI-exposure architectures construct different empirical treatments. They encode different technologies and occupational primitives, diverge in observable content and rankings, and draw identifying information from different occupations. Occupational mapping changes who enters the estimand, while common support does not eliminate differences in treatment-group membership or residual variation.

Despite that upstream divergence, every architecture produces a negative high-versus-low early-career employment-stock point estimate on literal common support. The confirmatory primary estimate also remains negative under every fixed-treatment occupation deletion. This is a robust aggregate direction for a specified tail contrast, not evidence of a monotone dose-response or a causal AI effect.

The agreement has sharp limits. One common-support interval includes zero under marginal inference, the predeclared simultaneous-band criterion does not establish all-six negativity, and mobility labels often disagree. Broad occupational assortativity explains nearly all of the realized excess mobility conflict over a stringent benchmark. The exploratory family decomposition places the negative conditional association more heavily on the Eloundou centroid, but its geometry is materially influenced by alpha and cannot rank families generally.

The durable conclusion is therefore neither that exposure choice is irrelevant nor that one index is uniquely correct. Alternative constructions can support the same aggregate employment tail contrast without being interchangeable for the rankings, comparisons, and economic interpretations attached to it. Robustness is statement-specific.

References

Acemoglu, Daron and Pascual Restrepo. 2019. "Automation and New Tasks: How Technology Displaces and Reinstates Labor." *Journal of Economic Perspectives* 33 (2): 3–30. 10.1257/jep.33.2.3.

Autor, David H. and David Dorn. 2013. "The Growth of Low-Skill Service Jobs and the Polarization of the US Labor Market." *American Economic Review* 103 (5): 1553–1597. 10.1257/aer.103.5.1553.

Autor, David H., Frank Levy, and Richard J. Murnane. 2003. "The Skill Content of Recent Technological Change: An Empirical Exploration." *Quarterly Journal of Economics* 118 (4): 1279–1333. 10.1162/003355303322552801.

Battaglia, Laura, Timothy Christensen, Stephen Hansen, and Szymon Sacher. 2024. "Inference for Regression with Variables Generated by AI or Machine Learning." CEPR Discussion Paper 19115, CEPR Press. https://cepr.org/publications/dp19115. Revised May 2025.

Bick, Alexander, Adam Blandin, and David J. Deming. 2024. "The Rapid Adoption of Generative AI." NBER Working Paper 32966, National Bureau of Economic Research. 10.3386/w32966. Revised February 2025.

Bick, Alexander, Adam Blandin, David J. Deming, and Tyler R. Schumacher. 2026. "What Work Does Generative AI Do?" NBER Working Paper 35677, National Bureau of Economic Research. 10.3386/w35677.

Brynjolfsson, Erik, Bharat Chandar, and Ruyu Chen. 2026. "Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence." , Stanford Digital Economy Lab. https://digitaleconomy.stanford.edu/publication/canaries-in-the-coal-mine-six-facts-about-the-recent-employment-effects-of-artificial-intelligence/. Revised August 12, 2026; original version published 2025.

Christensen, Timothy and Stephen Hansen. 2026. "Performing Valid Inference with AI/ML-Generated Covariates: A Guide for Empirical Practice." *AEA Papers and Proceedings* 116: 92–97. 10.1257/pandp.20261020.

Decancq, Koen and María Ana Lugo. 2013. "Weights in Multidimensional Indices of Wellbeing: An Overview." *Econometric Reviews* 32 (1): 7–34. 10.1080/07474938.2012.690641.

Dingel, Jonathan I. and Brent Neiman. 2020. "How Many Jobs Can Be Done at Home?" *Journal of Public Economics* 189: 104235. 10.1016/j.jpubeco.2020.104235.

Duan, Junting and Markus Pelger. 2026. "Inference with AI-Generated Covariates." NBER Working Paper 35481, National Bureau of Economic Research. 10.3386/w35481.

Eloundou, Tyna, Sam Manning, Pamela Mishkin, and Daniel Rock. 2024. "GPTs Are GPTs: Labor Market Impact Potential of LLMs." *Science* 384 (6702): 1306–1308. 10.1126/science.adj0998.

Emanuel, Natalia, Emma Harrington, and Amanda Pallais. 2026. "The Power of Proximity to Coworkers: Training for Tomorrow or Productivity Today?" *Quarterly Journal of Economics* 141 (3):

1825–1870. 10.1093/qje/qjag027.

Felten, Edward W., Manav Raj, and Robert Seamans. 2018. "A Method to Link Advances in Artificial Intelligence to Occupational Abilities." *AEA Papers and Proceedings* 108: 54–57. 10.1257/pandp.20181021.

Felten, Edward W., Manav Raj, and Robert Seamans. 2021. "Occupational, Industry, and Geographic Exposure to Artificial Intelligence: A Novel Dataset and Its Potential Uses." *Strategic Management Journal* 42 (12): 2195–2217. 10.1002/smj.3286.

Gimbel, Martha, Joshua Kendall, and Ryan Kulsakdinun. 2026. "Labor Market AI Exposure: What Do We Know?" https://budgetlab.yale.edu/research/labor-market-ai-exposure-what-do-we-know. The Budget Lab at Yale.

Hampole, Menaka, Dimitris Papanikolaou, Lawrence D. W. Schmidt, and Bryan Seegmiller. 2025. "Artificial Intelligence and the Labor Market." NBER Working Paper 33509, National Bureau of Economic Research. 10.3386/w33509. Revised September 2025.

Hu, Yingyao and Susanne M. Schennach. 2008. "Instrumental Variable Treatment of Nonclassical Measurement Error Models." *Econometrica* 76 (1): 195–216. 10.1111/j.0012-9682.2008.00823.x.

Humlum, Anders and Emilie Vestergaard. 2025. "Still Waters, Rapid Currents: Early Labor Market Transformation under Generative AI." NBER Working Paper 33777, National Bureau of Economic Research. 10.3386/w33777. Revised March 2026.

Kambourov, Gueorgui and Iourii Manovskii. 2008. "Rising Occupational and Industry Mobility in the United States: 1968–97." *International Economic Review* 49 (1): 41–79. 10.1111/j.1468-2354.2008.00473.x.

Ludwig, Jens, Sendhil Mullainathan, and Ashesh Rambachan. 2026. "Large Language Models: An Applied Econometric Framework." *Annual Review of Economics* 18: 283–316. 10.1146/annurev-economics-120925-105620.

Mellow, Wesley and Hal Sider. 1983. "Accuracy of Response in Labor Market Surveys: Evidence and Implications." *Journal of Labor Economics* 1 (4): 331–344. 10.1086/298019.

Merola, Rossana, Ekkehard Ernst, Daniel Samaan, R. Maria del Rio-Chanona, and Ole Teutloff. 2026. "Workers' Exposure to AI: What Indicators Tell Us—and What They Don't." Research brief, International Labour Organization. 10.54394/00033279.

Organisation for Economic Co-operation and Development. 2026. "The OECD AI Exposure Measure: Mapping the OECD AI Capability Indicators to Occupations." OECD Artificial Intelligence Papers 59, OECD Publishing. 10.1787/f3da0f0a-en.

Pulito, Giuseppe, Mariola Pytlikova, Sarah Schroeder, and Magnus Lodefalk. 2026. "Who Adopts AI? Evidence on Firms, Technologies and Workers." IZA Discussion Paper 18515, IZA Institute of Labor Economics. https://docs.iza.org/dp18515.pdf.

Romano, Joseph P. and Michael Wolf. 2005. "Stepwise Multiple Testing as Formalized Data Snooping." *Econometrica* 73 (4): 1237–1282. 10.1111/j.1468-0262.2005.00615.x.

U.S. Census Bureau. 2006. "Current Population Survey: Design and Methodology." Technical Paper 66, U.S. Census Bureau. https://www.census.gov/housing/hvs/files/tp-66.pdf.

U.S. Census Bureau. 2025. *Basic Current Population Survey Interviewer's Manual.* U.S. Census Bureau.

Webb, Michael. 2020. "The Impact of Artificial Intelligence on the Labor Market." , Stanford University. https://www.michaelwebb.co/webb_ai.pdf.

Yin, Michelle and Burhan Ogut. 2026. "Who Uses AI? Platform Selection and the Measurement of Occupational AI Exposure." Working Paper 2605.21743v2, arXiv. 10.48550/arXiv.2605.21743.

Yin, Michelle, Hoa Vu, and Claudia Persico. 2026. "How (un)Stable Are LLM Occupational Exposure Scores? Evidence from Multi-Model Replication." NBER Working Paper 35110, National Bureau of Economic Research. 10.3386/w35110.